

Section : Biology

Q.1 Which of the following statements are true with reference to homology or homologous organs?

(A) Homology indicates common ancestry.
(B) Whale and Cheetah share similarities in the pattern of the bones of the forelimbs.

(C) Vertebrate heart is an example of homologous organs.

(D) Thorn of *Bougainvillea* and tendrils of *cucurbita* represent homology.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.2 Match List I with List II

List I

List II

Technique/term

Features

(A) Amniocentesis

(I) Ligation of fallopian tube

(B) Maternal mortality rate

(II) Ligation of vas deferens

(C) Vasectomy

(III) Death rate

(D) Tubectomy

(IV) Testing of sex

Choose the correct options:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.3 Which of the following is incorrect with reference to drug abuse?

1. Cannabinoids affect cardiovascular system of the body.
2. Heroin is extracted from the latex of *Papaver somniferum*.
3. Nicotine is a very effective sedative and pain killer.
4. Excessive dosage of coca alkaloid causes hallucinations.

Options 1. 1

2. 2

3. 3

4. 4

Q.4 A small standing crop of phytoplankton supports a large standing crop of zooplankton. This can be correctly represented by:

1. Inverted pyramid of energy
2. Inverted pyramid of biomass
3. Upright pyramid of biomass
4. Sometimes upright and sometimes inverted pyramid of energy

Options 1. 1

2. 2

3. 3

4. 4

Q.5 ELISA is based on the principle of -

1. Antigen - antibody interaction
2. PCR
3. Radioactive molecule

4. Amount of DNA

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Match List-I with List-II

List-I

List-II

Product

Producer

(A) Citric Acid

(I) *Trichoderma polysporum*

(B) Ethanol

(II) *Monascus purpureus*

(C) Statins

(III) *Saccharomyces cerevisiae*

(D) Cyclosporin A (IV) *Aspergillus niger*

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Which one of the following options will express intermediate skin colour in an individual?

1. AABbCC
2. aabbcc
3. AaBbCc
4. aaBbcc

Options 1. 1

2. 2
3. 3
4. 4

Q.8 Which of the following are not involved in intra uterine devices?

- (A) Lippes Loop
- (B) LNG-20
- (C) Saheli
- (D) Implants

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (C) and (D) only
3. (A), (B), (C) and (D)
4. (C), (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.9 The process of copying genetic information from one strand of DNA into RNA is termed as

1. Replication
2. Translation
3. Transcription
4. Regulation

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Which one of the following guides the entry of the pollen tube into the embryo sac?

1. Antipodal Cells
2. Filiform apparatus
3. Micropyle
4. Synergids

Options 1. 1

2. 2
3. 3
4. 4

Q.11 Humification leads to :

1. Soil erosion
2. Soil conservation
3. Accumulation of humus
4. Accumulation of salts

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Which of the following steps are related to polymerase chain reaction (PCR)?

- (A) Extension
- (B) Annealing
- (C) Propagation
- (D) Denaturation

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.13 Match List-I with List-II

List-I

List-II

Name of the gene Encodes

- | | |
|---------|------------------------------|
| (A) 'i' | (I) Permease |
| (B) 'z' | (II) Repressor |
| (C) 'y' | (III) Transacetylase |
| (D) 'a' | (IV) β - Galactosidase |

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.14 If a man is eating fish as his food, which trophic level is occupied by him in the food chain?

1. First
2. Second
3. Third
4. Fourth

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Match List-I with List-II

List-I	List-II
Disease	Characteristic symptoms
(A) Malaria	(I) Inflammation of lower limb
(B) Filariasis	(II) Cycles of fever
(C) Ringworms	(III) Blood clots and excess mucus in stools
(D) Amoebiasis	(IV) Scaly lesions on nails

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Baculoviruses are pathogens that attack:

1. Insects
2. Roundworms
3. Molluscs
4. Birds

Options 1. 1

2. 2
3. 3
4. 4

Q.17 Which type of innate Immunity is exhibited by interferons secreted by virus infected cells to protect non infected cells?

1. Physical barriers
2. Cytokine barriers
3. Physiological barriers
4. Cellular barriers

Options 1. 1

2. 2
3. 3
4. 4

Q.18 *EcoRI*, a significant tool in rDNA technology is -

1. Bacteria
2. Plasmid
3. Enzyme
4. Purine

Options 1. 1

2. 2
3. 3
4. 4

Q.19 Which one of the following is opposite part of micropylar end of ovule.

1. Hilum
2. Chalaza
3. Funicle
4. Nucellus

Options 1. 1

2. 2
3. 3

Q.20 Which of the following disorders are the results of aneuploidy?

- (A) Haemophilia
- (B) Down's Syndrome
- (C) Thalassemia
- (D) Turner's Syndrome

Choose the correct answer from the options given below:

- 1. (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (A) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Which one of the following is not associated with the process of transcription in bacteria?

- 1. Rho factor
- 2. Methyl guanosine triphosphate
- 3. Sigma factor
- 4. DNA dependent RNA polymerase

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Genetically modified plants have been useful in:

- 1. Increasing post harvest losses
- 2. Decreasing crop yield
- 3. Making crops tolerant to stresses.
- 4. Decreasing efficiency of mineral usage by plants

Options 1. 1

2. 2

3. 3

4. 4

Q.23 The number of individuals in the reproductive age group is more than the number of individuals in the pre-reproductive age group, the shape of its age pyramid would reflect the growth status of the population as:

- 1. Expanding
- 2. Stable
- 3. Declining
- 4. Homeostasis

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Wildlife safari parks are example of-

- 1. *Ex-situ* Conservation
- 2. *In-situ* conservation
- 3. Biodiversity hot spots
- 4. Sacred grooves

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Arrange the following geological periods in their occurrence from latest to oldest order.

- (A) Triassic
- (B) Carboniferous

- (C) Tertiary
(D) Jurassic

Choose the correct answer from the options given below:

1. (D), (B), (C), (A)
2. (A), (B), (D), (C),
3. (B), (A), (D), (C)
4. (C), (D), (A), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Antibody molecule has:

1. Two peptide chain
2. Three peptide chain
3. Four peptide chain
4. One polypeptide chain

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Thorn of *Bougainvillea* and tendril of *Cucurbita* are:

1. Homologous structures
2. Analogous structures
3. Vestigial structures
4. Developing structures

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Transgenic animals are used to understand the contribution of genes in the development of diseases such as:

1. Cholera and typhoid
2. Elephantiasis and ringworm
3. Cancer and cystic fibrosis
4. Pneumonia and kala-azar

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Arrange the following events in the correct order pertaining to fertilization in the human reproductive system.

- (A) The blastocyst becomes embedded in the endometrium.
- (B) Finger-like projections appear on the trophoblast called chorionic villi.
- (C) The blastomeres are arranged into trophoblast and the inner cell mass.
- (D) The zygote divides mitotically and transforms into an embryo with 8-16 blastomeres, called a morula.

Choose the correct answer from the options given below.

1. (D), (C), (A), (B)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Of the incident solar radiation, what is the percentage of photosynthetically active radiation (PAR)?

1. 100%
2. Less than 50%

3. 1-5%
4. 2-10%

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Arrange the following groups of plants according to their appearance on earth.

- (A) Angiosperms
(B) Seed ferns
(C) Rhynia type plants
(D) Psilophyton

Choose the correct answer from the options given below:

1. (C), (D), (B), (A)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Arrange the given products formed during sewage treatment in correct sequence ?

- (A) Biogas
(B) Activated sludge
(C) Floccs
(D) Primary sludge

Choose the correct options given below

1. (A), (B), (C), (D)
2. (D), (C), (B), (A)
3. (C), (D), (B), (A)
4. (B), (A), (D), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.33 If a double-stranded DNA has 15% of adenine, find out the percent of cytosine in the DNA ?

1. 15%
2. 30%
3. 35%
4. 85%

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Arrange the given steps involved in gel electrophoresis used for separation of DNA fragments ?

- (A) Exposure to UV light
(B) Staining with ethidium bromide
(C) Moving of DNA fragments towards anode
(D) Elution

Choose the correct options given below

1. (A), (B), (C), (D)
2. (B), (A), (D), (C)
3. (A), (D), (B), (C)
4. (C), (B), (A), (D)

Options 1. 1

2. 2
3. 3

Q.35 Which one of the following structure is haploid (n) in relation to male reproductive system?

1. Secondary spermatocytes
2. Primary spermatocytes
3. Leydig cells
4. Sertoli cells

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Which of the following is the primary female sex organ?

1. Mammary glands
2. Uterus
3. Ovaries
4. Cervix

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Match List-I to List-II

List-I

List-II

Producer

Acid

- | | |
|----------------------------------|-------------------|
| (A) <i>Clostridium butylicum</i> | (I) Lactic acid |
| (B) <i>Aspergillus niger</i> | (II) Butyric acid |
| (C) <i>Acetobacter aceti</i> | (III) Citric acid |
| (D) <i>Lactobacillus</i> | (IV) Acetic acid |

Choose the correct options

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
3. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.38 The first isolated restriction endonuclease was:

1. *Hind II*
2. *EcoR I*
3. *BamH I*
4. *Pvu II*

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Choose the correct equation from the following options to show correct relationship between Gross Primary Productivity (GPP), Respirational Losses (R) and Net Primary Productivity (NPP).

1. $GPP = R / (NPP)^2$
2. $GPP = 2NPP + R$
3. $GPP = NPP - R$
4. $GPP - R = NPP$

Options 1. 1

2. 2
3. 3

Q.40 Match the List-I with List-II**List-I****List-II****Character****Recessive Trait**

(A) Flower Colour (I) Green

(B) Seed colour (II) Yellow

(C) Pod colour (III) Constricted

(D) Pod shape (IV) White

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

2. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.41 Read the passage carefully and answer the questions based on the passage:

Species diversity on earth is not uniformly distributed. It is generally highest in the tropics and decreases towards the poles. Earth's fossil history reveals the incidences of mass extinctions in the past. Earth's rich biodiversity is vital for the very survival of mankind. It is believed that communities with high diversity tend to be less variable and more productive. The reasons of conserving biodiversity are narrowly utilitarian, broadly utilitarian and ethical. Biodiversity conservation may be *in-situ* as well as *ex-situ*.

Which of the following is not included in in-situ conservation?

1. Zoological park

2. National park

3. Wild life sanctuary

4. Biosphere reserves

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Read the passage carefully and answer the questions based on the passage:

Species diversity on earth is not uniformly distributed. It is generally highest in the tropics and decreases towards the poles. Earth's fossil history reveals the incidences of mass extinctions in the past. Earth's rich biodiversity is vital for the very survival of mankind. It is believed that communities with high diversity tend to be less variable and more productive. The reasons of conserving biodiversity are narrowly utilitarian, broadly utilitarian and ethical. Biodiversity conservation may be *in-situ* as well as *ex-situ*.

Which of the following might not account for the greater biological diversity in the tropic region?

1. Frequent glaciations in the past

2. More solar energy available

3. Less seasonal and more constant and predictable

4. Undisturbed for million of years

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the passage carefully and answer the questions based on the passage:

Species diversity on earth is not uniformly distributed. It is generally highest in the tropics and decreases towards the poles. Earth's fossil history reveals the incidences of mass extinctions in the past. Earth's rich biodiversity is vital for the very survival of mankind. It is believed that communities with high diversity tend to be less variable and more productive. The reasons of conserving biodiversity are narrowly utilitarian, broadly utilitarian and ethical. Biodiversity conservation may be *in-situ* as well as *ex-situ*.

How many episodes of mass extinction of species have occurred since the origin and diversification of life on earth?

1. Two
2. Three
3. Five
4. Seven

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the passage carefully and answer the questions based on the passage:

Species diversity on earth is not uniformly distributed. It is generally highest in the tropics and decreases towards the poles. Earth's fossil history reveals the incidences of mass extinctions in the past. Earth's rich biodiversity is vital for the very survival of mankind. It is believed that communities with high diversity tend to be less variable and more productive. The reasons of conserving biodiversity are narrowly utilitarian, broadly utilitarian and ethical. Biodiversity conservation may be *in-situ* as well as *ex-situ*.

Which one of the following does not exhibit narrowly utilitarian argument for conserving biodiversity?

1. Construction materials
2. Pollination
3. Industrial products
4. Medicines

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the passage carefully and answer the questions based on the passage:

Species diversity on earth is not uniformly distributed. It is generally highest in the tropics and decreases towards the poles. Earth's fossil history reveals the incidences of mass extinctions in the past. Earth's rich biodiversity is vital for the very survival of mankind. It is believed that communities with high diversity tend to be less variable and more productive. The reasons of conserving biodiversity are narrowly utilitarian, broadly utilitarian and ethical. Biodiversity conservation may be *in-situ* as well as *ex-situ*.

Which of the following hot spots does not cover India's biodiversity regions?

1. Western Ghats-Sri Lanka
2. Amazon forests
3. Indo-Burma
4. Himalaya

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the questions based on the passage:

Pollen pistil interaction involves all events from the landing of pollen grains on the stigma until the pollen tube enters the embryo sac (when the pollen is compatible). When a pollen tube grows through the style and enters into the ovules, it finally discharges two male gametes in one of the

synergids. Syngamy and triple fusion are two fusion events occur in angiosperms. Thus, angiosperms exhibit double fertilization. The products of these fusions are the diploid zygote and triploid primary endosperm nucleus. Zygote develops into embryo and primary endosperm cell forms the endosperm tissue. The developing embryo passes through different stages before maturation.

With reference to reproduction in flowering plants, which one of the following is incorrect?

1. Endosperm develops into seed
2. Ovary develops into fruit
3. Ploidy of PEN is $3n$
4. Syngamy is the fusion of male and female gamete

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the questions based on the passage:

Pollen pistil interaction involves all events from the landing of pollen grains on the stigma until the pollen tube enters the embryo sac (when the pollen is compatible). When a pollen tube grows through the style and enters into the ovules, it finally discharges two male gametes in one of the synergids. Syngamy and triple fusion are two fusion events occur in angiosperms. Thus, angiosperms exhibit double fertilization. The products of these fusions are the diploid zygote and triploid primary endosperm nucleus. Zygote develops into embryo and primary endosperm cell forms the endosperm tissue. The developing embryo passes through different stages before maturation.

If there are 38 chromosomes in the zygote, how many chromosomes will be there in its haploid egg cell?

1. 38
2. 57
3. 19
4. 76

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the questions based on the passage:

Pollen pistil interaction involves all events from the landing of pollen grains on the stigma until the pollen tube enters the embryo sac (when the pollen is compatible). When a pollen tube grows through the style and enters into the ovules, it finally discharges two male gametes in one of the synergids. Syngamy and triple fusion are two fusion events occur in angiosperms. Thus, angiosperms exhibit double fertilization. The products of these fusions are the diploid zygote and triploid primary endosperm nucleus. Zygote develops into embryo and primary endosperm cell forms the endosperm tissue. The developing embryo passes through different stages before maturation.

The coconut water from tender coconut, a good source of nutrition is nothing but:

1. Free-nuclear endosperm
2. Synergids
3. Antipodal cells
4. Scutellum

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the questions based on the passage:

Pollen pistil interaction involves all events from the landing of pollen grains on the stigma until the pollen tube enters the embryo sac (when the

pollen is compatible). When a pollen tube grows through the style and enters into the ovules, it finally discharges two male gametes in one of the synergids. Syngamy and triple fusion are two fusion events occur in angiosperms. Thus, angiosperms exhibit double fertilization. The products of these fusions are the diploid zygote and triploid primary endosperm nucleus. Zygote develops into embryo and primary endosperm cell forms the endosperm tissue. The developing embryo passes through different stages before maturation.

Which of the following is not a stage of growing embryo in dicotyledon plants?

1. Heart-shaped
2. Globular
3. Proembryo
4. Epiblast

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the questions based on the passage:

Pollen pistil interaction involves all events from the landing of pollen grains on the stigma until the pollen tube enters the embryo sac (when the pollen is compatible). When a pollen tube grows through the style and enters into the ovules, it finally discharges two male gametes in one of the synergids. Syngamy and triple fusion are two fusion events occur in angiosperms. Thus, angiosperms exhibit double fertilization. The products of these fusions are the diploid zygote and triploid primary endosperm nucleus. Zygote develops into embryo and primary endosperm cell forms the endosperm tissue. The developing embryo passes through different stages before maturation.

Formation of diploid zygote is a result of:

1. Emasculation
2. Triple fusion
3. Syngamy
4. Bagging

Options 1. 1

2. 2
3. 3
4. 4